Monitoring Amazon Document DB serverless

The key metrics to track the Amazon Document DB Serverless for the performance and cost optimization are provided below:

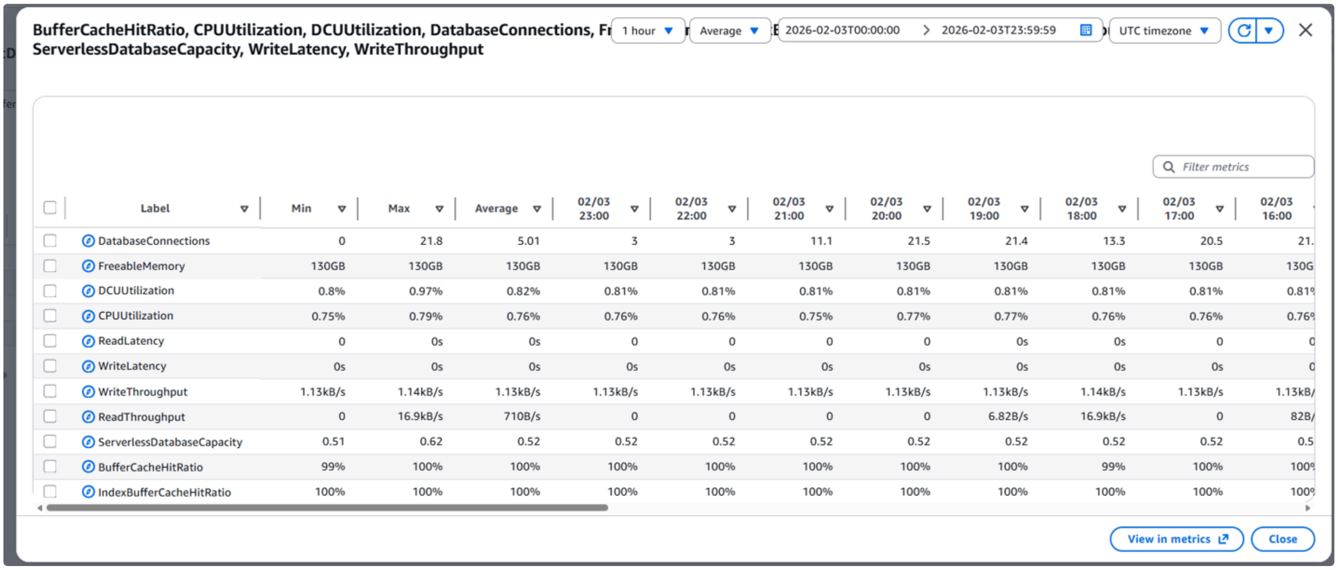
Key Metrics to track

Category	Metric Name	Purpose	What to track?
Capacity & Utilization	DCUUtilization	Percentage of the maximum DCUs being used. High values indicate you may need to increase the max DCUs.	• Shows how much of your maximum DCU capacity is being used. • If consistently near 100%, increase **max DCUs**. • If consistently very low, consider lowering **min DCUs** to save cost.
Capacity & Utilization	ServerlessDatabaseCapacity	As an instance-level metric, it reports the number of DCUs represented by the current instance capacity. As a cluster-level metric, it represents the average of the `ServerlessDatabaseCapacity` values of all the Document DB serverless instances in the cluster.	• Actual DCUs consumed at any point. • Compare against min/max settings to see if scaling is effective.
Performance	CPUUtilization	Percentage of CPU used by the cluster. Amazon DocumentDB monitors this value automatically and scales up your serverless	• High CPU during peak loads may indicate insufficient DCUs

		instance when the instance consistently uses a high proportion of its CPU capacity.	
Performance	FreeableMemory	This value represents the amount of unused memory that is available when the DocumentDB serverless instance is scaled to its maximum capacity. Low memory can cause performance issues.	• Low memory during peak loads may indicate insufficient DCUs
Performance	DatabaseConnections	Number of active client connections; spikes may indicate scaling needs.	• Spikes in connections without corresponding DCU scaling could mean limits are too tight.
Performance	BufferCacheHitRatio	Indicates how often data requests are served from the buffer cache instead of disk.	• A high ratio means most reads are from memory, which is good for performance. • A low ratio suggests frequent disk reads, which can slow queries.
Performance	IndexBufferCacheHitRatio	similar to BufferCacheHitRatio but specifically for index pages.	• A high ratio means index lookups are efficient because they're cached. • A low ratio could indicate insufficient memory or poorly optimized indexes.
Latency & Throughput	ReadLatency / WriteLatency	Average time for read/write operations	• Increasing latency or throttling during high utilization suggests max DCUs are too low.
Latency & Throughput	ReadThroughput / WriteThroughput	Bytes read/written per second.	• Increasing latency or throttling during high

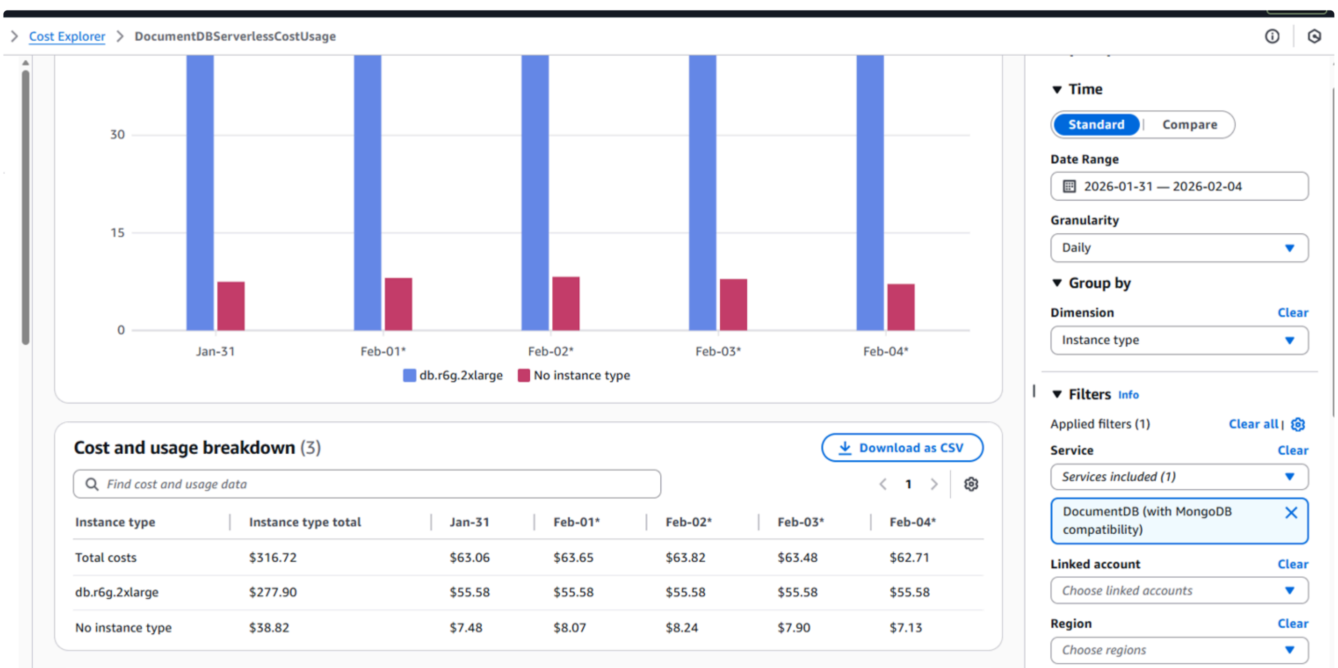
				utilization suggests max DCUs are too low.

Sample Cloudwatch Metrics during Idle period



Label	Min	Max	Average	02/03 23:00	02/03 22:00	02/03 21:00	02/03 20:00	02/03 19:00	02/03 18:00	02/03 17:00	02/03 16:00
DatabaseConnections	0	21.8	5.01	3	3	11.1	21.5	21.4	13.3	20.5	21.
FreeableMemory	130GB	130GB	130GB	130GB	130GB	130GB	130GB	130GB	130GB	130GB	130G
DCUUtilization	0.8%	0.97%	0.82%	0.81%	0.81%	0.81%	0.81%	0.81%	0.81%	0.81%	0.81
CPUUtilization	0.75%	0.79%	0.76%	0.76%	0.76%	0.75%	0.77%	0.77%	0.76%	0.76%	0.76
ReadLatency	0	0s	0s	0	0	0	0	0s	0s	0	0
WriteLatency	0s	0s	0s	0s	0s	0s	0s	0s	0s	0s	0
WriteThroughput	1.13kB/s	1.14kB/s	1.13kB/s	1.13kB/s	1.13kB/s	1.13kB/s	1.13kB/s	1.13kB/s	1.14kB/s	1.13kB/s	1.13kB/
ReadThroughput	0	16.9kB/s	710B/s	0	0	0	0	6.82B/s	16.9kB/s	0	82B/
ServerlessDatabaseCapacity	0.51	0.62	0.52	0.52	0.52	0.52	0.52	0.52	0.52	0.52	0.5
BufferCacheHitRatio	99%	100%	100%	100%	100%	100%	100%	100%	99%	100%	100
IndexBufferCacheHitRatio	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100

Sample Cost Usage during Idle period



Instance type	Instance type total	Jan-31	Feb-01*	Feb-02*	Feb-03*	Feb-04*
Total costs	$316.72	$63.06	$63.65	$63.82	$63.48	$62.71
db.r6g.2xlarge	$277.90	$55.58	$55.58	$55.58	$55.58	$55.58
No instance type	$38.82	$7.48	$8.07	$8.24	$7.90	$7.13

Best Practices

DocumentDB Serverless automatically scales between **MinCapacity** and **MaxCapacity** based on workload. Proper configuration and usage patterns help reduce unnecessary costs and ensure resources scale down promptly when idle.

1. **Configure Min/Max DCU Appropriately**
   - Set **MinCapacity** as low as possible (e.g., 0.5 DCU) for idle periods.
   - Define **MaxCapacity** based on peak workload requirements to avoid over-provisioning.
2. **Close Idle Connections**
   - Terminate unused sessions promptly.
   - Implement **connection pooling with timeouts** to prevent the cluster from staying scaled up.
3. **Disable Features That Prevent Scale-Down**
   - Turn off **Performance Insights** or similar monitoring features if not required.
   - Avoid continuous health checks or background queries that simulate activity.
4. **Monitor & Tune**
   - Use **CloudWatch metrics** to track DCU usage and identify patterns.
   - Adjust Min/Max DCU settings based on observed workload trends.
5. **Validate Scale Behavior**
   - Test how quickly the cluster scales down after workloads complete.
   - Ensure no background tasks (e.g., index builds, TTL cleanup) are preventing scale-down.
6. **Validate Scale Behavior**
   - **Set alarms** on thresholds (e.g., DCUUtilization >80%, memory <500MB, latency spikes) to prompt that configuration needs to be reviewed and adjusted.

Dev Document DB Serverless Cluster metrics

Attached the consolidated report on the testing performed on the Dev Document DB serverless cluster.

Serverless_Dev...cs.xlsx
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